INTRODUCTION TO APPLIED NUMERICAL METHODS

Ordinary Differential Equations: Euler's Method

Lang Yuan, Associate Professor

UNIVERSITY OF
South Carolina

College of Engineering
and Computing

BACKGROUND

Variables	Annotation
m	Parachutist mass
t	Time
c	Drag coefficient
g	gravity
v	Velocity

Newton's Second Law

$$\frac{dv}{dt} = a = \frac{F}{m} = \frac{F_D + F_U}{m} = \frac{mg - cv}{m} = g - \frac{c}{m}v$$

Ordinary Differential Equations (ODEs)

$$\boxed{\frac{dv}{dt} = g - \frac{c}{m}v}$$

- An independent variable (t)
- A dependent variable (v)
- LHS: Derivative of dependent variable (v) with respect to independent variable (t)
- RHS: A function (t, v, or constant, …)

UofSC Engineering and Computing

BACKGROUND

Fundamental Laws in ODEs

Law	Mathematical Expression	Variables and Parameters
Newton's second law of motion	$\frac{dv}{dt} = \frac{F}{m}$	Velocity (v), force (F), and mass (m)
Fourier's heat law	$q = -k' \frac{dT}{dx}$	Heat flux (q), thermal conductivity (k') and temperature (T)
Fick's law of diffusion	$J = -D \frac{dc}{dx}$	Mass flux (J), diffusion coefficient (D), and concentration (c)
Faraday's law (voltage drop across an inductor)	$\Delta V_L = L \frac{di}{dt}$	Voltage drop (ΔV_L), inductance (L), and current (i)

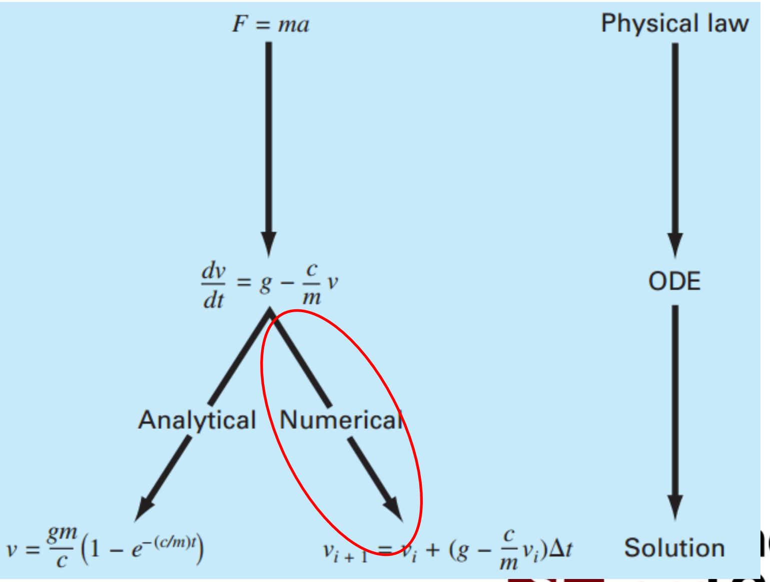
Analytical Solution:
<u>Continuous solution within the entire domain</u>

$$y = F(x, y)$$

Numerical Solution:
<u>Discrete solution (data points) within the entire domain</u>

$$(x_i, y_i)$$

Solution Methods



RUNGE-KUTTA METHOD

$$\frac{dy}{dx} = f(x, y)$$

to compute

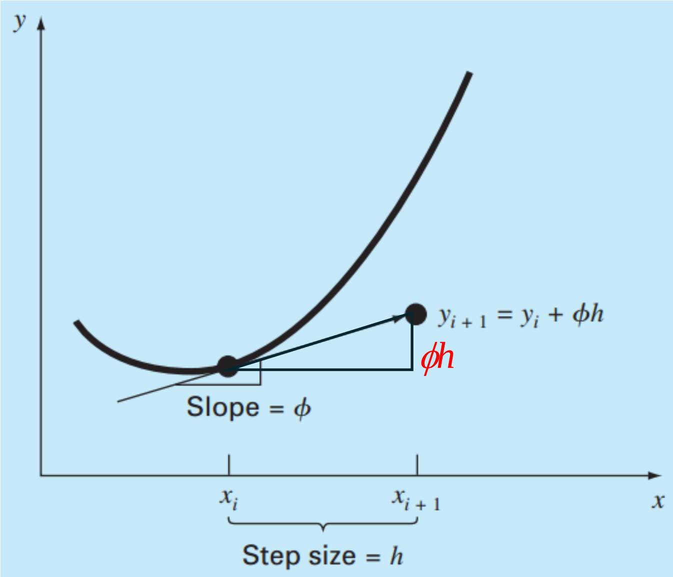
$$\frac{y_{i+1} - y_i}{x_{i+1} - x_i} = f(x, y)$$

$$y_{i+1} = y_i + f(x, y)(x_{i+1} - x_i) = y_i + f(x, y)h$$

$\phi = f(x, y)$

$$\boxed{y_{i+1} = y_i + \phi h}$$

New value = old value + slope × step size



ϕ:

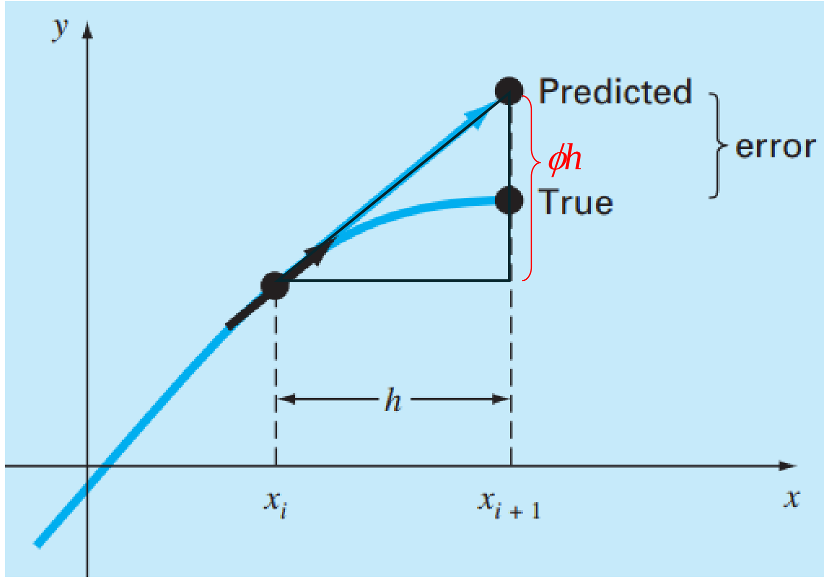
- The slope estimate
- Used to extrapolate from an old value y_i to a new value y_{i+1} over a distance h
- The different methods to estimate the slope ϕ lead to different numerical solution methods
- All generally called "*Runge-Kutta*" methods

EULER'S METHOD

$$\boxed{y_{i+1} = y_i + \phi h}$$

New value = old value + slope × step size

$$\boxed{\phi = f(x_i, y_i)}$$



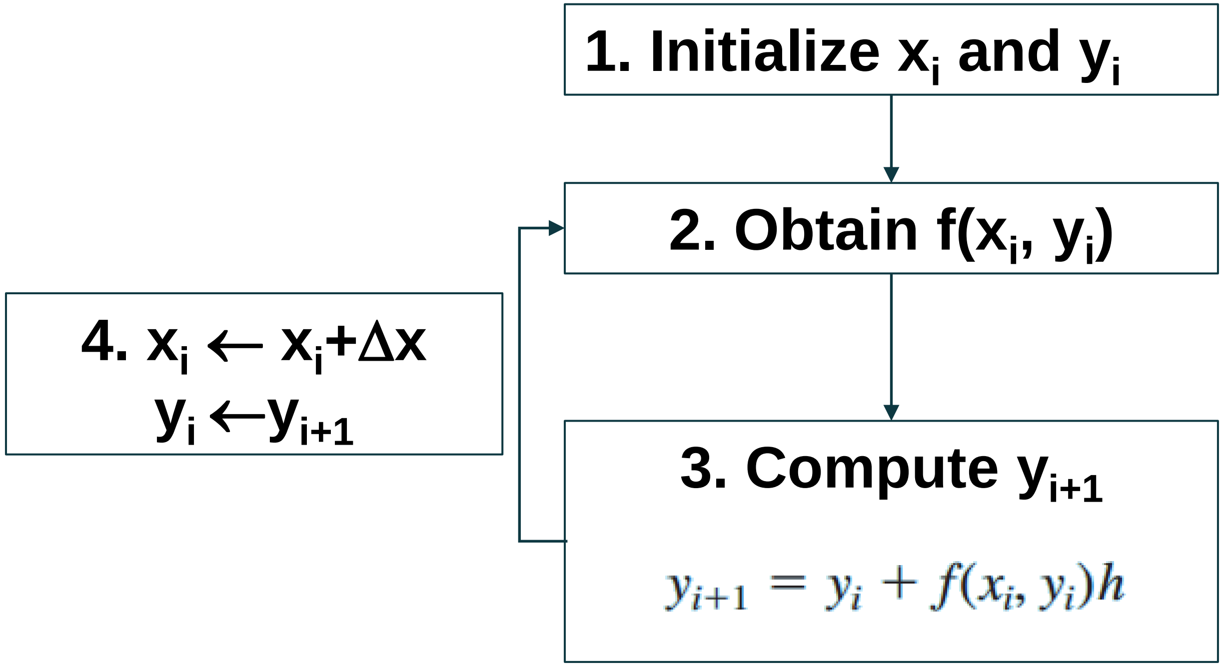
ϕ* direct estimate of the slope at *x_i

$$y_{i+1} = y_i + f(x_i, y_i)h$$

- Prediction based on the first-order Taylor expansion
- A new value of y is predicted using the slope (the first derivative at the original value of x) to extrapolate linearly over the step size *h*

EULER'S METHOD

Flow Chart for Euler's solution to ODE



AN EXAMPLE

Using the Euler's method to numerically integrate

$$\frac{dy}{dx} = f(x,y) = -2x^3 + 12x^2 - 20x + 8.5$$

From x = 0 and x = 4 with a step size of 0.5 with the initial condition x = 0 and y = 1

Solution:

The exact solution is: $y = -0.5x^4 + 4x^3 - 10x^2 + 8.5x + 1$

The numerical solution:

@ $x_i = 0$ @ $x_i = 0$ & $y = 1$

@ $x_{i+1} = 0.5$ ➡ $y_{i+1} = y_i + f(x_i, y_i)h$ ➡ $y(0.5) = y(0) + f(0, 1)0.5$

$$y(0) = 1 \quad f(0,1) = -2(0)^3 + 12(0)^2 - 20(0) + 8.5 = 8.5$$
$$y(0.5) = 1.0 + 8.5(0.5) = 5.25$$

@ $x_{i+1} = 1$

$$\begin{aligned} y(1) &= y(0.5) + f(0.5, 5.25)0.5 \\ &= 5.25 + [-2(0.5)^3 + 12(0.5)^2 - 20(0.5) + 8.5]0.5 \\ &= 5.875 \end{aligned}$$

AN EXAMPLE

Using the Euler's method to numerically integrate

$$\frac{dy}{dx} = f(x, y) = -2x^3 + 12x^2 - 20x + 8.5$$

From *x* = 0 and *x* = 4 with a step size of 0.5 with the initial condition x = 0 and y = 1

Solution:

The exact solution is: $y = -0.5x^4 + 4x^3 - 10x^2 + 8.5x + 1$

@ x = 0.5 $\quad y_t = -0.5(0.5)^4 + 4(0.5)^3 - 10(0.5)^2 + 8.5(0.5) + 1 = 3.21875$

$$\varepsilon_t = \left|\frac{3.21875 - 5.25}{3.21875}\right| \times 100\% = 63.1\%$$

@ x = 1 $\quad y_t = -0.5 + 4 - 10 + 8.5 + 1 = 3$

$$\varepsilon_t = \left|\frac{3 - 5.875}{3}\right| \times 100\% = 95.8\%$$

AN EXAMPLE

Using the Euler's method to numerically integrate

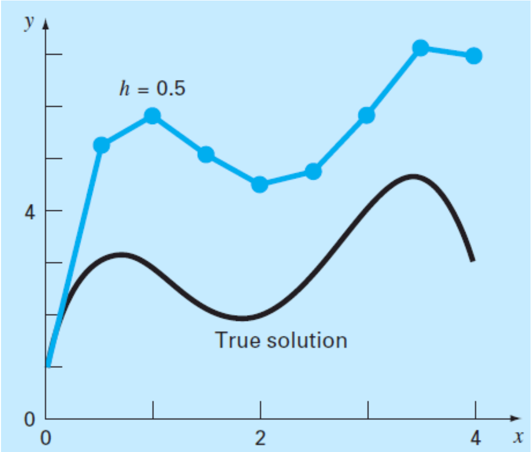
$$\frac{dy}{dx} = f(x,y) = -2x^3 + 12x^2 - 20x + 8.5$$

From x = 0 and x = 4 with a step size of 0.5 with the initial condition x = 0 and y = 1

Solution:

x	y_{true}	y_{Euler}	Percent Relative Error
0.0	1.00000	1.00000	
0.5	3.21875	5.25000	−63.1
1.0	3.00000	5.87500	−95.8
1.5	2.21875	5.12500	−131.0
2.0	2.00000	4.50000	−125.0
2.5	2.71875	4.75000	−74.7
3.0	4.00000	5.87500	−46.9
3.5	4.71875	7.12500	−51.0
4.0	3.00000	7.00000	−133.3

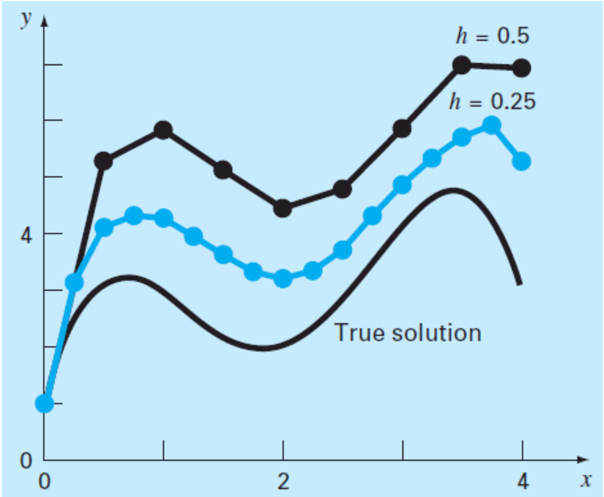
- The computation captures the general trend of the true solution
- The error is considerable
- **This error can be reduced by using a smaller step size**

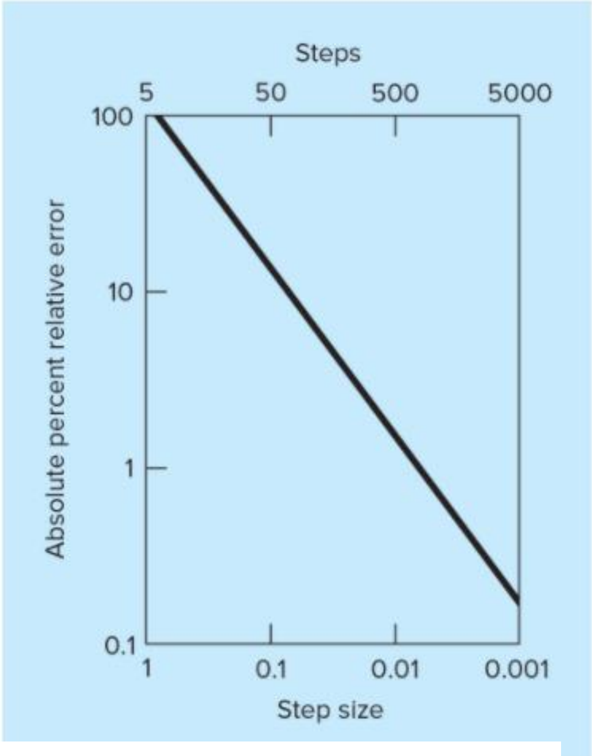


IMPROVING EULER'S METHOD

$$\boxed{y_{i+1} = y_i + \underbrace{f(x_i, y_i)h + \frac{f'(x_i, y_i)}{2!}h^2 + \cdots + \frac{f^{(n-1)}(x_i, y_i)}{n!}h^n + O(h^{n+1})}_{\text{Truncation errors}}}$$

- This error can be reduced by using a smaller step size
- Normally the true value is not known *a priori*. Euler's method using a number of different step sizes needs to be evaluated to examine the effect of the step size
- The method will provide error-free predictions if the underlying function (that is, the solution of the differential equation) is linear, i.e., y = y(x) is linear, and f(x,y) is a constant





Effect of step size at x=5